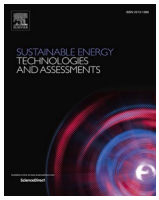
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Exergy and energy analysis of sensible heat storage based double pass hybrid solar air heater

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ABSTRACT

This research work discusses the exergy and energy analysis of an innovative double pass solar air heater with sensible heat storage. The system is evaluated in three different operating conditions; namely, natural convection mode, induced forced convection mode, and induced forced convection mode with a reflector. Performance analysis in all operating conditions evaluated and the best operating condition selected to apply crop drying and space heating application. Metco material and aluminum scrap mixture were used as sensible heat storage material below the bottom plate of the system. Experiments were conducted on three consecutive days for the different operating conditions. Average energy and exergy efficiency under natural convection mode is 11.47% and 2%, while 56% and 10.43% for a system operated under induced forced convection mode; however, 86.19% and 17.617% for a system operated under induced forced convection mode with a reflector. The third condition (system under induced forced convection mode with a reflector) has superior heat performance than the other two cases. Therefore, the proposed system is highly suitable for low-temperature applications.

Introduction

Environmental pollution has increased in the last 20 years. Poisonous gas emissions like CO, NOx, etc have increased linearly due to the overuse of nonrenewable energy sources. Therefore, regular planning and research are needed to replace the application of exhaustible energy resources. Renewable energy has a broad scope of research and can replace nonrenewable sources of energy in the future. Solar technology and its applications are the current topics of research and innovation. Initiative and planning are being carried forward to harness maximum energy from solar panels and solar power plants to meet the energy demand and reduce energy stress on coal power plants.

One of the eco-friendly applications of solar technology is the solar air heater (SAH). It is a cost-effective renewable device used to heat air for proper conditioning in commercial and industrial buildings. Solar air heater (SAH) is application of solar thermal technology that absorbs solar radiation through an absorber plate and heating the air flowing through the ducts. SAH uses solar energy for space heating, crop drying and other industrial usages. There are various types of solar air heaters designed depending upon the requirement. A double pass solar air heater (DPSAH) is one of such kind with superior thermal performance compared to single-pass SAH. DPSAH utilizes heat losses from the absorber plate to heat the air passed from the bottom collector [1]. However, to increase efficiency, some design criteria should be maintained, including the absorber area, the absorptance of the absorber, and airflow.

Many researchers have worked towards the concept of DPSAH. Bahrehmand et al. [2] studied the exergy and energy analysis of different types of solar air collectors using tin absorber plate. Experimental research concludes that the double glass cover SAHs have better performance than single glass cover SAH. Research also found that triangular fins in the collector are more energy-efficient than rectangular fins. Velmurugan et al. [3] analyzed that triple-pass SAH gives better exergy and energy performance than single pass and DPSAH. Ghiami and Ghiami [4] conducted a comparative study based on energy and exergy analyses of a baffled solar air heater with latent storage collector. Rise in air temperature, exergy gain, and efficiency of a SAH

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Nomenclature

Q_A	Total heat (W)
Q_u	Heat utilized (W)
Q_{loss}	Heat loss (W)
Q_{st}	Incident heat energy stored in the heater area (W)
m	Mass flow rate (MFR), kg/s
C_p	Specific Heat in constant pressure (J/kg.K)
T_{out}	Outlet Temperature (^{o} C)
T_{in}	Inlet Temperature (^{o} C)
V_{ai}	Velocity of air (m/s)
S	Area of the duct (m^2)
A_p	Area of Absorber Plate (m^2)
I_i	Solar radiation (W/m^2)
U_{loss}	Overall heat loss coefficient [$W/(m^2K)$]
T_p	Absorber plate temperature (°C)
T_a	Ambient Temperature (°C)
U_e	Side heat loss coefficient [$W/(m^2K)$]
U_b	Bottom heat loss coefficient [$W/(m^2K)$]
U_t	Top heat loss coefficient [$W/(m^2K)$]
$h_{r.Pg}$	Radiative heat transfer coefficient ($Wm^{-2}K^{-1}$)
$h_{c.Pg}$	Convective heat transfer coefficient ($Wm^{-2}K^{-1}$)
$h_{r.ga}$	Radiative heat transfer coefficient due to wind velocity ($Wm^{-2}K^{-1}$)
h_w	External radiative and convective heat transfer coefficient due to velocity of wind ($Wm^{-2}K^{-1}$)
Nu	Nusselt Number
Ra	Reynolds number
Pr	Prandtl number
t	Temperature (°C)
T_g	Temperature of glass plate (°C)
T_a	Ambient Temperature (°C)
ε_p	Emissivity of the plate
V	Velocity of air (m/s)
T_{sky}	Temperature of the external air at the surrounding (°C)
L_1	Length of Solar air heater (m)
L_2	Width of solar air heater (m)
L_3	Height of the solar air heater (m)
I_T	Solar radiation (W/m^2)
g	Gravitation constant ($m^3\ kg^{-1}\ s^{-2}$)
ΔT	Temperature difference of air, (°C)
L	Length of the solar air heater (m)
ν	Kinematic viscosity of air (m^2/s)
D_h	Characteristic distance (m)
Ex_{in}	Exergy at entrance (J)
Ex_d	Exergy destroyed (J)
Ex_{out}	Exergy at outlet (J)
Ex_{st}	Exergy stored (J)
Ex_{los}	Exergy lost (J)
P_{in}	Air Pressure inside solar air heater [$kg/(m^2)$]
P_a	Atmospheric Pressure [$kg/(m^2)$])
I	incident solar radiation (W/m^2)
T_s	Mean temperature of the absorber plate (°C)
r	Distribution of solar radiation
A_c	Area of the collector (m^2)
$Ex_{d(p-s)}$	Exergy destroyed due to pressure drop (J)
η_0	Efficiency (%)
$Ex_{d(\Delta P)}$	Exergy destroyed from absorber plate to surrounding (J)
ΔP	Pressure drop [$kg/(mS^2)$]
$Ex_{d(p-f)}$	Exergy destroyed from absorber plate surface to flowing air (J)
W_X	Uncertainty due to anomalous system
$W_{Temperature}$	Error due to temperature measuring instrument
$W_{thermometer}$	Error due to thermometer
$W_{reading}$	Reading error
W_{Wind}	Error due to wind speed
$W_{anemometer}$	Error due to wind speed and pressure
W_{SR}	System deformities error
$W_{Pyranometer}$	Error in solar irradiance on a planar surface
$W_{Experimental}$	Experimental error

Greek Symbol

α	Absorptivity
ρ	Density of air (kg/m^3)
ε_p	Emissivity of absorber plate
ε_g	Emissivity of glass surface
λ	Insulation thermal conductivity (W/mK)
λ_e	Insulation thermal conductivity at the side (W/mK)
δ_b	Insulation thickness at the bottom (mm)
δ_e	Insulation thickness at the side (mm)
σ	Stefan–Boltzmann Constant ($W/m^2\ K^4$)
τ	Transmissivity
η_o	Overall thermal efficiency (%)
μ	Air viscosity (Ns/m^2)
β	Thermal expansion coefficient

depend on the mass flow rate (MFR) of air, solar radiation intensity, and the absorber plate's surface geometry. Chamoli et al. [5] analytically found that V-down pierced baffles in absorber plate can result in better exergy performance by increasing solar heat absorption and dissipation potential. Khadraoui et al. [6] analyzed the use of paraffin wax as a PCM (phase change material) in SAH. The use of paraffin as PCM can increase the daily (including night) efficiency of a SAH up to 33%. Chabane et al. [7] studied the efficiency of single-pass SAH using longitudinal fins by varying the MFR. Zukowski [8] studied the flow characteristics and efficiency of a novel microjet SAH. Developed device achieved heat conversion efficiency ranging from 66% to 90%, while a single glass cover can improve efficiency from 16% to 24%. Saxena et al. [9] conducted an experiment to study free and forced convection in SAH using a mixture of granular carbon and desert sand as PCM coupled with sensible thermal storage materials. Comparison is drawn between two similar sets of operations in natural convective flow. Others operating in forced convective flow stated that 19–20.7% thermal efficiency was achieved in natural convection and 53–80% thermal efficiency was achieved in forced convection. Siddiqui et al. [10] did a study to determine the role of SAH in design and performance enhancement in the water desalination process by the solar humidification-dehumidification system. Experiment was conducted in Medina, Saudi Arabia on two experimental setups, one without SAH and two electric heaters, while the second was set up with one electric heater and one SAH. The increased daily productivity rate and simultaneous cost decrease in the second experimental set up one with an electric heater and SAH placed before the humidification chamber. Pramanik et al. [11] did an experimental study to evaluate the performance of DPSAH. Better designed SAH, a dc fan and selective coating on the absorber plate were used. Average thermal efficiency and outlet temperature increased, and the difference between the experimental and theoretical values lies between 1.2% and 9.7%. Kalaiarasi et al. [12] designed a type of SAH with absorber plates made up of copper tubes filled with synthetic oil. Exergy analysis of the absorber cum heat storage is performed. Maximum exergy and energy efficiency obtained was 50–59% for higher MFR 0.026 kg/s. Moreover, Edalatpour et al. [13] did exergy and energy analysis of a double glazed SAH, which had paraffin wax as PCM. Experimental study was done on the same set up for three days of summer to evaluate daily exergy and

energy analysis. Daily energy efficiency of the system varied from 58% to 69%, while the exergy efficiency ranged from 14% to 26% in latitude, 37° 28′ N and longitude, 57° 20′ E. A double pass finned SAH studied by Fudholi et al. [14] at different MFRs and solar radiation. Study reveals that variation in MFRs and solar radiation affects energy efficiency, exergy efficiency and improvement potential for solar air heater. Srinivas et al. [15] experimented on photovoltaic SAH where the absorber plate is partially covered by a photovoltaic cell. It had been observed that the overall energy (thermal and electrical) efficiency varied between 29 and 37%, whereas the exergy efficiency varied between 14 and 17%. Abuşka [16] experimented on SAH with attached conical fins at the absorber plate to increase its efficiency and performance. Energy efficiency increased up to 74.6% (MFR.1 kg/s) and exergy up to 19.3% (0.04 kg/s). Heydari et al. [17] found out that DPSAH with helical channeling is 14.7% more efficient than simple ducts. Salih et al. [18] discovered that increased airflow rate results in delaying the melting point during charging of PCM. However, the inverse case was observed during the discharge since the airflow rate was inversely related to freezing time. Satyender et al. [19] did experimental work with converging finned wire mesh packed bed SAH. Experiment concluded that the converging sections improve the convective heat transfer enhancing thermal and thermohydraulic efficiency to 93% and 80%. Rhee et al. [20] studied transpired, corrugated and simple flat plate collectors having Re = 4000–16,000. The corrugated and transpired configuration had better efficiency with optimal pressure drop. Prasad and Sah [21] did an experimental study on SAH collector by modifying it with reflective mirrors to concentrate the solar radiation and circular ribs created for turbulence. The result showed that for Re = 4000–16,000 as solar radiation increased to 40% while performance increased up to 90% when reflective mirror combined with turbulator were used. Manjunath et al. [22] did numerical study of SAH by using wave profiled absorber plate. For Re = 4000–24,000. the average thermal efficiency increased by 12.5% while MFR reduced. Thakur et al. [23] did numerical study on SAH by modifying the absorber plate by using hyperbolic rib profile. Optimum rib performance achieved at Re 6000, roughness 1 mm height and 10 mm pitch. Study concluded to have improvement in thermo-hydraulic performance of SAH. Jaurker et al. [24] stated that the maximum heat transfer occurred at 6 relative roughness pitch. Nu improved by 2.7 times and friction factor rose up to 3.6 times whereas pressure loss reduced through the grooves in the absorber plate and rectangular ribs. Satyender et al. [25] stated that 49 °C outlet air temperature achieved for optimum MFR 0.023 kg/s for double pass converging finned wire mesh pack SAH. Tyagi et al. [26] stated that DPSAH has maximum thermal efficiency up to 33% when aluminum metal used in absorber plate. Pfister et al. [27] did a study of six grids assembly corrugated in a triangular fashion so that the air traverse 16 times through SAH with MFR 0.025–0.095.80% conversion efficiency was achieved for 0.06 kg/s MFR. Krishnananth and Murugavel [28] did experiment in SAH to study the effect of aluminum capsules and paraffin-waxes material. Thermal performance increased due to property of the material to act as a thermal storage. Anup et al. [29] stated that the heat transfer through modified SAH increased with 2.46% increase in Nusselt number and 1.78 times increase in friction factor for the normal duct at 60° twist angle of twisted roughness. Baissi et al. [30] done study on rectangular channel artificially roughed with two longitudinally curved delta-shaped baffles with perforation and without perforations SAH. Highest thermal enhancement factor for perforated configuration (2.26%) while 2.21% for non-perforated configuration. Filiz et al. [31] did study on DPSAH by modifying absorber plate connecting it with aluminum cans. MFR was in the range of 0.03–0.05 kg/s, helped aluminum cans to reduce the dead zones in the collector by better airflow and turbulence on both side of the absorber plate. Jha et al. [32] did mathematical modelling of DPSAH in parallel mode heat transfer for space air conditioning application. Mathematical model predicted 3–7 °C temperature rise. Rajneesh et al. [33] did geometrical modification of the absorber triangular cross section duct with forward facing chamfered rectangular ribs having Re 4000–18,000. For 1.5 rib aspect ratio 2.64 times Nu increased to 17,000 while Re value was 4000. Binguang et al. [34] did study of spiral shaped flow path of the air inside the duct with MFR0.0146–0.0250. Spiral type SAH had efficiency and heat loss coefficient higher in comparison to conventional SAH with inlet and ambient temperature difference of 5 °C and 3% efficiency reduction.

In view of the above literature survey, it is noticed that a lot of experimental and numerical works have already been done to analyze the performance of the double pass solar air heater. However, no literature was found on the energy and exergy analysis of double pass solar air heater with sensible thermal storage assisted with reflector for the composite climate. Hence the motivation of the present study is to design and develop double pass solar air heater with sensible thermal storage coupled with steel reflector for composite climate. A double pass solar air heater was fabricated for experimentation to achieve these objectives. Observed parameters have been used for performance analysis in terms of energy and exergy analysis. Energy efficient, environmentally friendly and sustainable double pass solar air heater is manufactured with locally available materials. Experiments were conducted on three consecutive days under different modes (natural mode, induced forced mode and induced forced mode coupled with reflector). In the second and third sections, energy and exergy analysis had performed based on experimental data. Proposed system under induced forced mode coupled with reflector shows superior results in both analyses. At last, the thermal efficiency of the proposed system is compared with previous competitive literature. The result of the proposed system is found to be comparable with other systems.

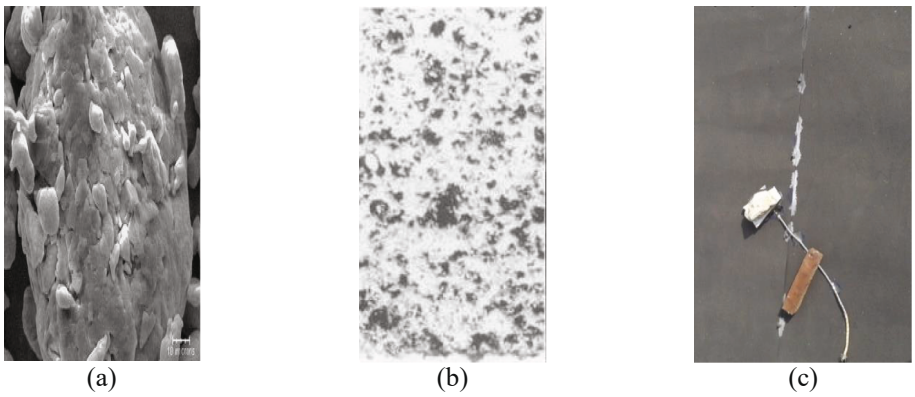
Material and methods

Experimental setup and procedure

A double pass solar air heater with sensible heat storage (DPSAHSHC) has been designed, developed, and experimented with in different conditions in the present study (Figs. 1–4). Fig. 1 shows the microscopic view of Metco material and its coating on the absorber plate. Dark surface having a high absorptance, and hence collector absorber is blackened if not naturally dark. Black surface would absorb over 95% of the incident solar radiation. Also, to avoid heat loss from the air heater wall, ceramic wool type insulation material having 20 mm thickness is used. Sunlight exposed collector led to useful heat towards air and some heat losses to the surrounding. Cumulative heat losses to the surrounding area are in three ways i.e. side, top and edge.

Fig. 2 represents the various section views of the proposed system. System is kept at tilt angle of the location (23°) and south-facing to obtain maximum solar radiation. Absorber plate of the DPSAHSHC is made up of a copper sheet with 2 mm thickness. Absorber surface is coated with a black metco material (Aluminum silicon-boron nitride powder with high absorptivity and low corrosion resistance). Copper or steel material has limitations since it would require a selective coating to maximize the absorption of solar energy and minimize the radiation emitted by the plate. Metco material with aluminum scrap (20 kg) is used as thermal storage material in DPSAHSHC. A 50 mm thick layer of Metco material with aluminum scrap is spread at the bottom inner wall of the SAH. In the natural mode, air comes in the down vent and get exhaust through upper vent. Various items involved in the manufacture of the proposed system and its embodied energy are given in Table 1. Embodied energy is the energy required to manufacture the proposed system from extraction to processing to final construction. There are twelve different components used in the proposed system. Among all, the polycarbonate sheet's contribution is the highest both by weight and embodied energy.

An 18-volt dc fan is fitted at the outlet vent to maintain air circulation inside the proposed system. DC fan can also be put at the inlet vent of the system; however, the number of air exchanges per minute will be



(a) (b) (c)

Fig. 1. (a) Microscopic view of metco material, (b) normal view of metco material, (c) Metco coating on the absorber plate.

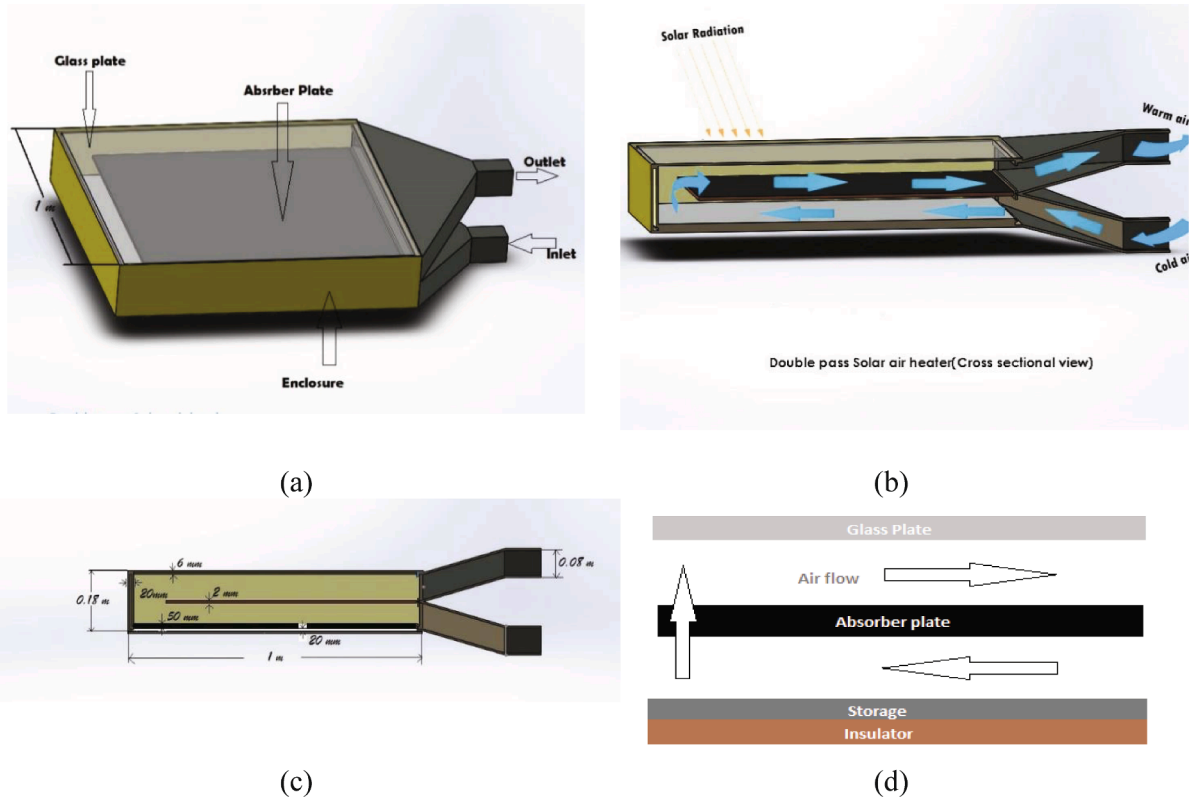


Fig. 2. Sectional view of double pass solar air heater with sensible thermal storage concept.

less than the outlet vent. Due to this reason, in this study, exhaust fan is kept at the outlet vent. A solar panel (single crystalline) with a 40-watt power supply assisted with solar charge controller with battery backup is used to power up dc fan. Due to this arrangement, DC fan get steady power supply despite the fluctuation in solar radiation. Fig. 3 is the representation of the air circulation in the natural and induced forced mode of convection. Due to the application of exhaust fan at the outlet vent, the number of air exchanges in the system under induced forced mode is much higher than the natural mode.

Fig. 4(a-e) represents a detailed experimental setup of the proposed system. Top layer of the proposed system is made of toughen glass to transmit solar radiation with minimum thermal loss to the absorber plate. Thickness of the toughened glass is 3 mm, and its surface area is 1 $\times$ 1 m^2. Outer side and bottom body are made of 4 mm thick polycarbonate sheet. Polycarbonates sheet is robust, flexible, and can reduce heat losses. The whole system is mounted on an iron stand.

DPSAHSHC collector consists of two channels, the first channel framed between the glass cover and absorber plate and the second channel formed under the absorber plate and sensible thermal storage bed. Thermal storage material is kept on the bottom of the system. Sides and bottom of the proposed system are insulated with ceramic wool. Area of outlet/inlet vent is 0.0484 m^2. Both vents are in square shape. Space between the glass cover and top absorber plate is 5 cm, while space between the top absorber plate and bottom layer is 3 cm. A polished stainless steel reflector is used in proposed system to receive more solar radiation. Size of reflector is 0.25 m^2 and framed in mild steel iron structure.

Experiments were conducted on three consecutive days starting from 7th - 9th April 2019 with a different configuration of air flow and reflector:

i. 1st day (7th April): On the first day of experiment, proposed system has been experimented in natural convection mode.

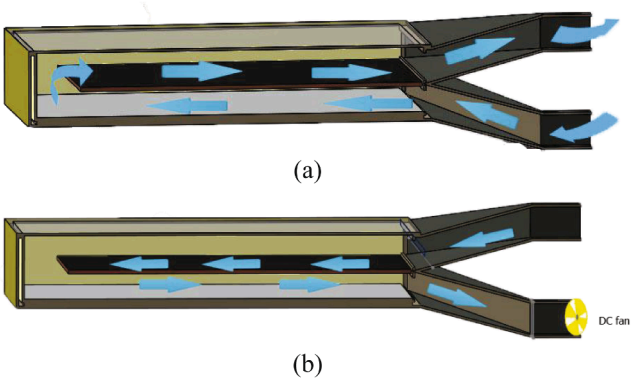


Fig. 3. Air flow arrangement in double pass solar air heater with sensible thermal storage (a) Natural convection mode (b) Induced force convection mode.

ii. 2nd day (8th April): On the second day of the experiment, the fan is fitted in the bottom duct of the system. Thus, airflow allowed entering through the upper duct to perform induced forced convection in the SAH.
iii. 3rd day (9th April): On third day, the experiment was conducted under induced forced convection mode with reflector.

(A) Uncertainty analysis

Every measuring instrument has limitations; none of the measuring devices is 100% accurate. Uncertainty analysis is intended to indicate the possible error in measurement by the instruments. The uncertainty analysis is being calculated with the help of Eqs. (29)–(33). External Uncertainty Percentage Error is given Table 2 [49].

$$W_X = \sqrt{\left(\frac{\partial X}{\partial x_1}W_1\right)^2 + \left(\frac{\partial X}{\partial x_2}W_2\right)^2 + \left(\frac{\partial X}{\partial x_3}W_3\right)^2 + \ldots\ldots + \left(\frac{\partial X}{\partial x_n}W_n\right)^2} \quad (29)$$

$$W_{Temperature} = \sqrt{(W_{thermometer})^2 + (W_{reading})^2} \quad (30)$$

$$W_{Wind} = \sqrt{(W_{anemometer})^2 + (W_{reading})^2} \quad (31)$$

$$W_{SR} = \sqrt{(W_{Pyranometer})^2 + (W_{reading})^2} \quad (32)$$

$$W_{Experimental} = \sqrt{(W_{Wind})^2 + (W_{Temperature})^2 + + (W_{SR})^2} \quad (33)$$

By solving all these equations (Eqs. (30)–(33)) with the help of Table 2, total uncertainty is found to be ±0.2004% which is in actable range.

Methodology

A. Energy analysis

Energy analysis of the proposed system was conducted with the help of experimental data. This analysis is based on the conservation of energy principle. For the present system, the following energy balance equations are used:

$$Q_A = Q_u + Q_{loss} + Q_{st} \quad (1)$$

Now each term can be expressed in the following ways:

Q_u is expressed as the total amount of heat utilized and which is mathematically expressed as follows [35]:

$$Q_u = \dot{m}C_p(T_{out} - T_{in}) \quad (2)$$

where, m is the MFR of the air inside the system. This can be evaluated by the following expression [35]:

$$\dot{m} = \rho V_{ai} S \quad (3)$$

In this system, the primary source of energy is solar radiation. Net solar radiation absorbed by the absorber plate is denoted by Q_A which is evaluated by the following expression [35]:

$$Q_A = A_p(\alpha\tau)I_i \quad (4)$$

In the solar collector, some heat loss occurs on the bottom side and top surface of the system. Therefore, the total amount of heat loss is calculated by the following expression [35]:

$$Q_{loss} = U_{loss}A_p(\bar{T}_p - T_a) \quad (5)$$

U_{loss} is total heat losses coefficient which include losses from the top, bottom and side. Mathematically it is expressed as follow [36]:

$$U_{loss} = U_e + U_b + U_t \quad (6)$$

U_t is calculated as [36]:

$$U_t = \left[1/(h_{r,Pg} + h_{c,Pg}) + 1/(h_{r,ga} + h_w)\right]^{-1} \quad (7)$$

Flow is found to be laminar inside the solar collector. Hence, the Nusselt number is presented as Eq. (8) [20,37-39]:

$$Nu = 0.68 + \frac{(0.67 \times Ra^{1/4})}{\left(1 + (0.492/Pr)^{9/16}\right)^{4/9}} \; [Ra \leqslant 10^9] \quad (8)$$

$$h_{c,pg} = Nu(\lambda/t) \quad (9)$$

$$h_{r,pg} = \left(\frac{\sigma\left(T_g^2 + T_P^2\right)(T_g + T_p)}{\left(\frac{1}{\varepsilon_p} + \frac{1}{\varepsilon_p} - 1\right)}\right) \quad (10)$$

External radiative and convective heat transfer due to the wind velocity evaluated by Hottle and Woertz [20,40] is represented in Eq. (11) and Eq. (12), respectively.

$$h_w = 3.86V + 5.67 \quad (11)$$

$$h_{r,ga} = \varepsilon_g\sigma(T_g^2 + T_{sky}^2)(T_g + T_{sky}) \quad (12)$$

where T_{sky} calculated by the following Eq. (13) [20,40]:

$$T_{sky} = 0.0552T_a^{1.5} \quad (13)$$

Heat loss coefficient for bottom and sidewalls of the SAH are represented in Eq. (14) and Eq. (15) respectively [41].

$$U_b = \lambda_i/\delta_b \quad (14)$$

$$U_e = ((L_1 + L_2)L_3\lambda_i)/(L_1L_2\delta_e) \quad (15)$$

where L_1L_2, and L_3 are length, width and height of the SAH respectively.

According to the first law of thermodynamics, the thermal efficiency of the SAH is the ratio between the useful energy (Q_u) and the solar radiation received on the collector. Thermal efficiency can be calculated by the Eq. (16) [42]:

$$\eta_{Thermal} = Q_u/I_TA_p \quad (16)$$

Dimensionless parameters referred above are given by Eq. (17), Eq. (18) and Eq. (19) [43]:

$$Ra = \frac{g\beta\Delta TL^3}{\alpha v} \quad (17)$$

$$Pr = \frac{v}{\alpha} \quad (18)$$

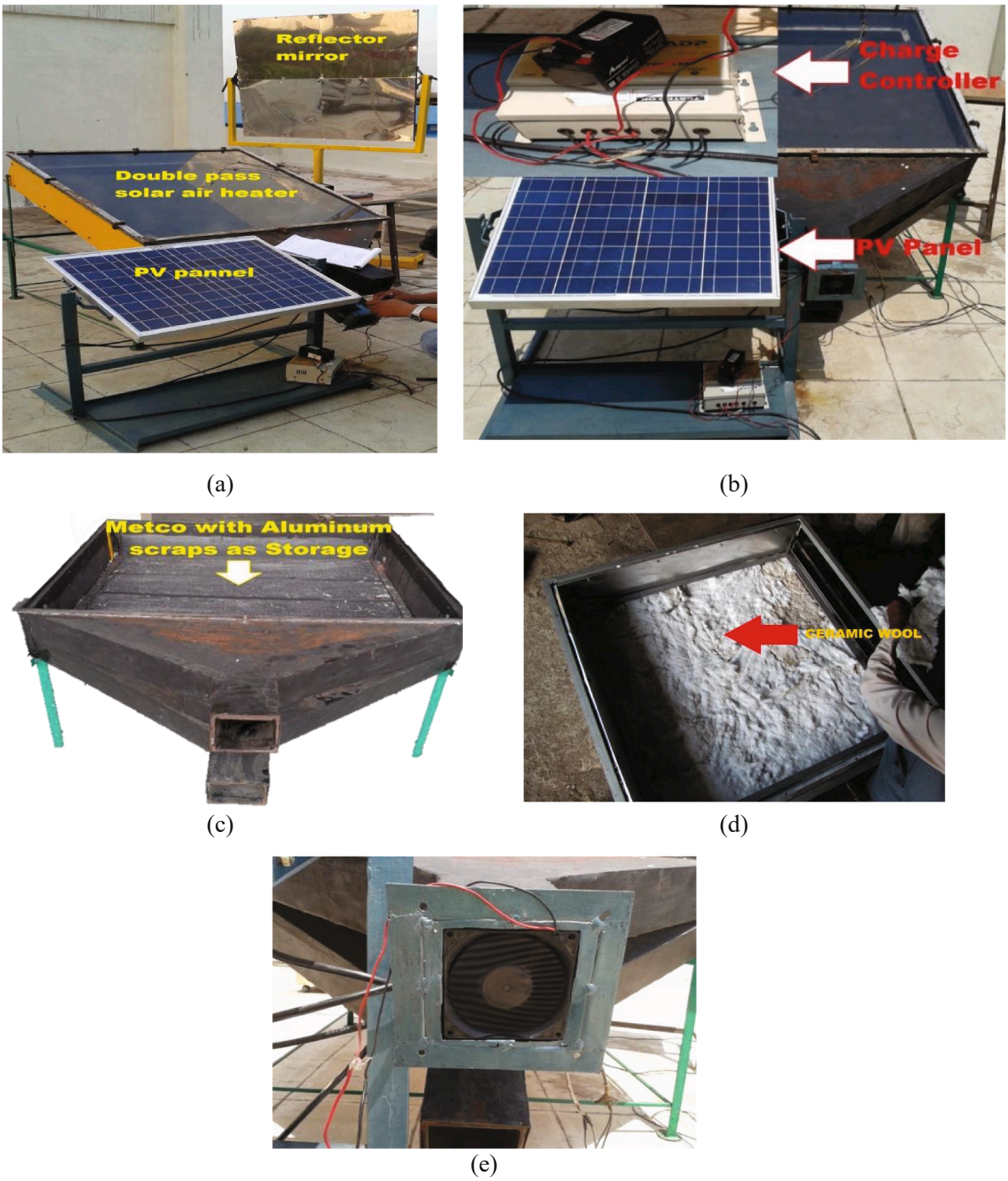


Fig. 4. (a-e): Experimental set up of DPSAH.

$$Re = \frac{VD_h}{v} \tag{19}$$

B. Exergy analysis of the SAH

These are the assumption considered for exergy analysis of the proposed system based on second law of thermodynamics:

i. Steady-state flow operation.
ii. Airflow is an ideal gas and with constant physical property.
iii. Negligible humidity of the air.
iv. No chemical reaction.
v. Although the system is installed with an inclination angle of about 23°, potential and kinetic energy differences are considered negligible.

Exergy balance equation is being developed based on second law of thermodynamics. Exergy at the inlet of the SAH is given as Eq. (20):

$$Ex_{in} = Ex_d + Ex_{out} + Ex_{st} + Ex_{loss} \tag{20}$$

where Ex_{in}, $Ex_d Ex_{out} Ex_{st}$ and Ex_{loss} are the inlet, destroyed, outlet, stored and lost exergy, respectively.

Exergy for inlet airflow is calculated by Eq. (21) [44]:

Table 1
Embodied energy analysis for fabrication of SAH [50].

S. No.	Material	Quantity	Embodied Energy Coefficient	Total (kWh)
1	Glass	5.40 kg	7.28 (kWh/kg)	39.312
2	Polycarbonate sheet	15.60 kg	10.20 (kWh/kg)	159.12
3	Black coating	0.75 m^2	0.28 (kWh/m^2)	0.21
4	Copper Sheet	0.30 kg	19.44 (kWh/kg)	5.832
5	Metco Material	12.00 kg	26.76 (kWh/kg)	321.12
6	Aluminium Scrap	8.00 kg	18.98 (kWh/kg)	151.84
7	Ceramic Wool	1.00 kg	4.05 (kWh/kg)	4.05
8	DC fan			
	(I)Plastic	0.120	19.44 (kWh/kg)	2.332
	(II) Copper wire	0.050	19.61 (kWh/kg)	0.9805
9	Polycrystalline solar cell	0.059 m^2	1130.60 (kWh/m^2)	66.7054
10	Battery			0.046
11	Solar charge controller			0.033
12	Stainless steel	1 kg	55.55(kWh/m^2)	55.55
13	Iron	05 kg	6.94(kWh/m^2)	34.70
	Total Embodied Energy			**841.8309**

Table 2
External uncertainty percentage error

S. No.	Measurement Error	Value
1	Absorber temperature	0.10
2	Air temperature	0.10
3	Relative humidity	0.10
4	Moisture evaporation	0.50
5	Voltage	0.01
6	Current	0.01
7	Wind velocity	0.01
8	Plate Temperature	0.10
Total Percent External Uncertainty		**0.93**

$$Ex_{in}\,(\text{flowing air}) = \dot{m}C_P\left(T_{in} - T_a\left(1 + ln\left(\frac{T_{in}}{T_a}\right)\right)\right) + \dot{m}RT_a ln\left(\frac{P_{in}}{P_a}\right) \quad (21)$$

Exergy at outlet airflow is expressed as Eq. (22) [44]:

$$Ex_{out}(\text{flowingair}) = -\dot{m}C_P(T_{out} - T_a(1 + ln(\frac{T_{out}}{T_a}))) - \dot{m}RT_a ln(\frac{P_{out}}{P_a}) \quad (22)$$

Exergy for absorbed solar radiation can be calculated as Eq. (23) [45]:

$$Ex_{in}\,(\text{solar radiation}) = lr\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \left(\frac{T_a}{T_s}\right)^4\right] \quad (23)$$

where $T_s = 5777$ K.

Lost exergy, Ex_{loss} can be indicated as Eq. (24) [46]:

$$Ex_{loss} = -U_{loss}A_c\left[\left(T_P - T_a\right) - (T_a + 273)ln\frac{(T_P + 273)}{(T_a + 273)}\right] \quad (24)$$

Exergy destruction Ex_d is dependent upon three different terms as:

i. Temperature difference between the absorber plate and sun (shown in Eq. (25)) [47]
ii. Pressure drop at duct as shown in Eq. (26) [47]
iii. Heat transfer to the flowing air from absorber plate to the surface as shown in Eq. (27) [47]

$$Ex_{d(p-s)} = -\eta_0 I_T A_c T_a((\frac{1}{T_p}) - (\frac{1}{T_s})) \quad (25)$$

$$Ex_{d(\Delta P)} = -(\Delta P T_a/\rho)\left(T_a ln\left(\frac{T_{out}}{T_a}\right)\right)\Big/(T_{out} - T_{in}) \quad (26)$$

$$Ex_{d(p-f)} = -\dot{m}C_p T_a\left(\left(ln\left(\frac{T_{out}}{T_{in}}\right)\right) - \left(\frac{T_{out} - T_{in}}{T_p}\right)\right) \quad (27)$$

Daily exergy efficiency of the SAH can be calculated as Eq. (28) [48]:

$$\eta_{Exergy} = (\int Ex_{out}/\int Ex_{in}) \times 100\% \quad (28)$$

Result and discussion

Experiments evaluate the performance of a double pass solar heater in three different modes, namely natural convection, induced forced convection and induced forced convection with reflector mode. Exergy and energy performance of the proposed system are evaluated. The different measuring devices used to generate the data and obtain satisfactory results are listed in Table 3.

Effect of incident solar radiation on temperature difference, thermal and exergy efficiency

Incident solar radiation is the only energy source of the proposed system during three days of experimentation from 7th to 9th April 2019. Average solar radiation noted as 673.33 W/m^2, 671.11 W/m^2 and 665.55 W/$m^{2,}$ respectively. In natural convection conditions, the solar radiation at 9:00 AM was 400 W/$m^{2,}$ and the outlet temperature measured was 45 °C. However, the maximum solar radiation found was 950 W/m^2 (at 1:00 PM), and the maximum outlet temperature was 60 °C on that same day. On the second day and third day of the experiment, the solar radiation at 9:00 AM was 410 W/m^2 and 420 W/m^2 and the outlet temperature was 32 °C and 38.2 °C, respectively.

In contrast, the maximum solar radiation observed was 945 W/m^2 and 940 W/m^2 and the maximum outlet temperature was 58.2 °C and 73.2 °C, respectively. Fig. 5(a-c) shows the variation of inlet duct temperature, outlet duct temperature and solar radiation vs time. Higher solar radiation increased the differences between the inlet and outlet temperature and useful energy. Fig. 6 also depicts that the increase in solar radiation leads to increased thermal efficiency and exergy efficiency for all three experiment days. Thus, efficiency of solar air heater depends upon the solar irradiance and area exposed to solar radiation. An increase in solar radiation leads to an increase in temperature, thus reducing the heat loss through the system, thereby increasing the thermal and exergy efficiency. Intensity of solar radiation affects the collector thermal performance, where higher insolation and inlet temperature resulting in higher thermal efficiency.

Effect of temperature difference on thermal and exergy efficiency

Thermal efficiency is the amount of useful heat energy collected from solar radiation striking on the absorber surface. Exergy efficiency is the adequacy of a system comparative with its performance in reversible conditions. It is characterized as the ratio of the thermal effectiveness of a system compared with a reversible variant of the system. Air temperature that passes through the SAH is determined by subtracting air

Table 3
Technical instruments and measurement device.

Device	Quantity	Model/Producer	Range	Accuracy
Air blower	2	RS555/Themisto	0–500 RPM	±5 RPM
Anemometer	1	Lutron/AM4202	0.4–30 m/s	±0.1 ms^{-1}
Solar Power Meter	1	CEM/LA-1017	0e1999 W/m^2	±10 W/m^2
Thermocouples	8	K-type/Jain Automation	−200 to 1200 °C	±0.5 °C
Data logger	1	Testo/176T4	−195 to 1200 °C	±3 °C

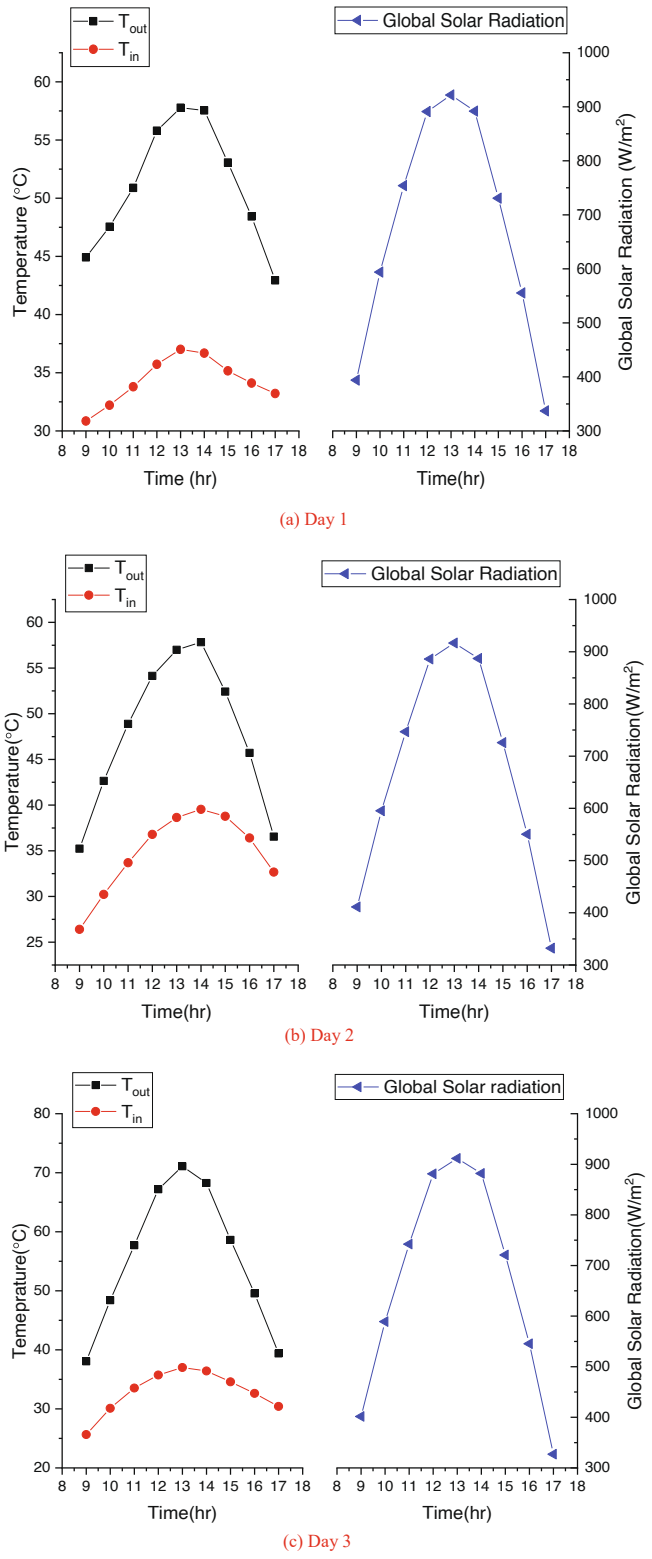


Fig. 5. (a-c): Comparison variation of inlet temperature, outlet temperature and solar radiation with time.

outlet duct temperature with inlet duct temperature, i.e., $\Delta T = T_{out} - T_{in}$. Temperature difference depends upon the operating mode of the DPSAH. The DPSAH under natural convection mode has an average temperature difference of 13.45 °C.

In contrast, when operated in induced forced convection mode with and without reflector, the same SAH has an average temperature

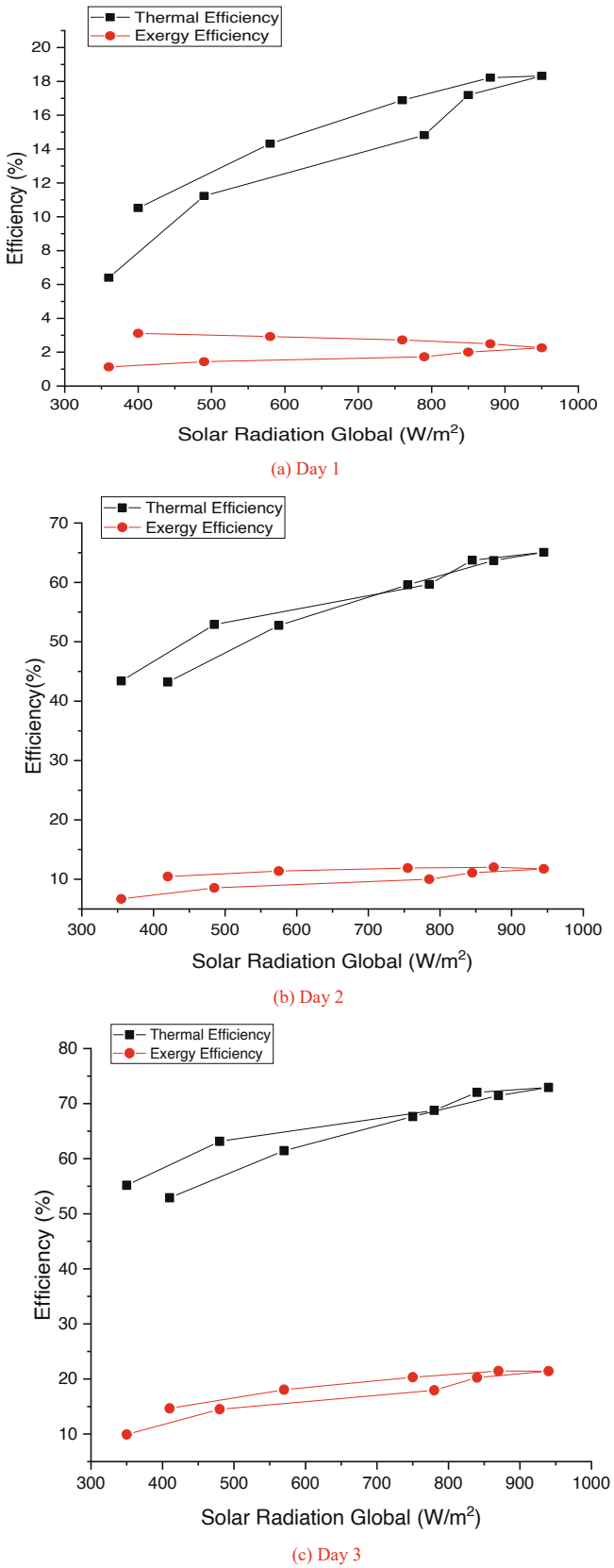


Fig. 6. Effect of solar radiation on thermal efficiency and exergy efficiency.

difference of 22.38 °C and 16.37 °C, respectively. Temperature difference is a major impact on thermal and exergy efficiency. Average thermal efficiency was 56% and 65.05%, respectively, for the 2nd and 3rd-day experiment with the same average MFR of 0.02 kg/s. Average exergy efficiency has been recorded as 13.21% and 16.17%, respectively, for the 2nd and 3rd day experiment. Fig. 7(a-c) compares thermal efficiency and temperature rise on SAH with variation in time. Thermal efficiency increased with temperature difference and further decreases with decreased temperature. Air flowing through the collector carries heat from the sun, naturally raising the air. Thus, cold outside air is heated and delivered, further improving the thermal efficiency. Thermal storage improves the performance of the system by storing and conserving energy to be further utilized by SAH.

Effect of dc fan

The dc fan is used to carry out forced convection in DPSAH. It results in more MFR through the duct of SAH. In the natural mode of operation, no dc fan was required. Hence the average MFR was 0.005 kg/s. It has been observed that both the exergy and thermal efficiency are least when no dc fan is used. In the natural convection mode of operation, the average thermal efficiency was 14.21%, and the exergy efficiency was observed as 3.22%. On 2nd day, fan is fitted at the bottom outlet duct of the DPSAH, resulting in an induced forced air with an average MFR of 0.02 kg/s from the upper duct to the bottom. Induced flow resulted in an average thermal and exergic efficiency of 56% and 13.21%, respectively, on the same SAH configuration. Figs. 8 and 9 show the comparison of thermal and exergy efficiency variation with time in all conditions of natural and induced forced convection mode. Exergy efficiency depends on the inlet and outlet temperature since energy

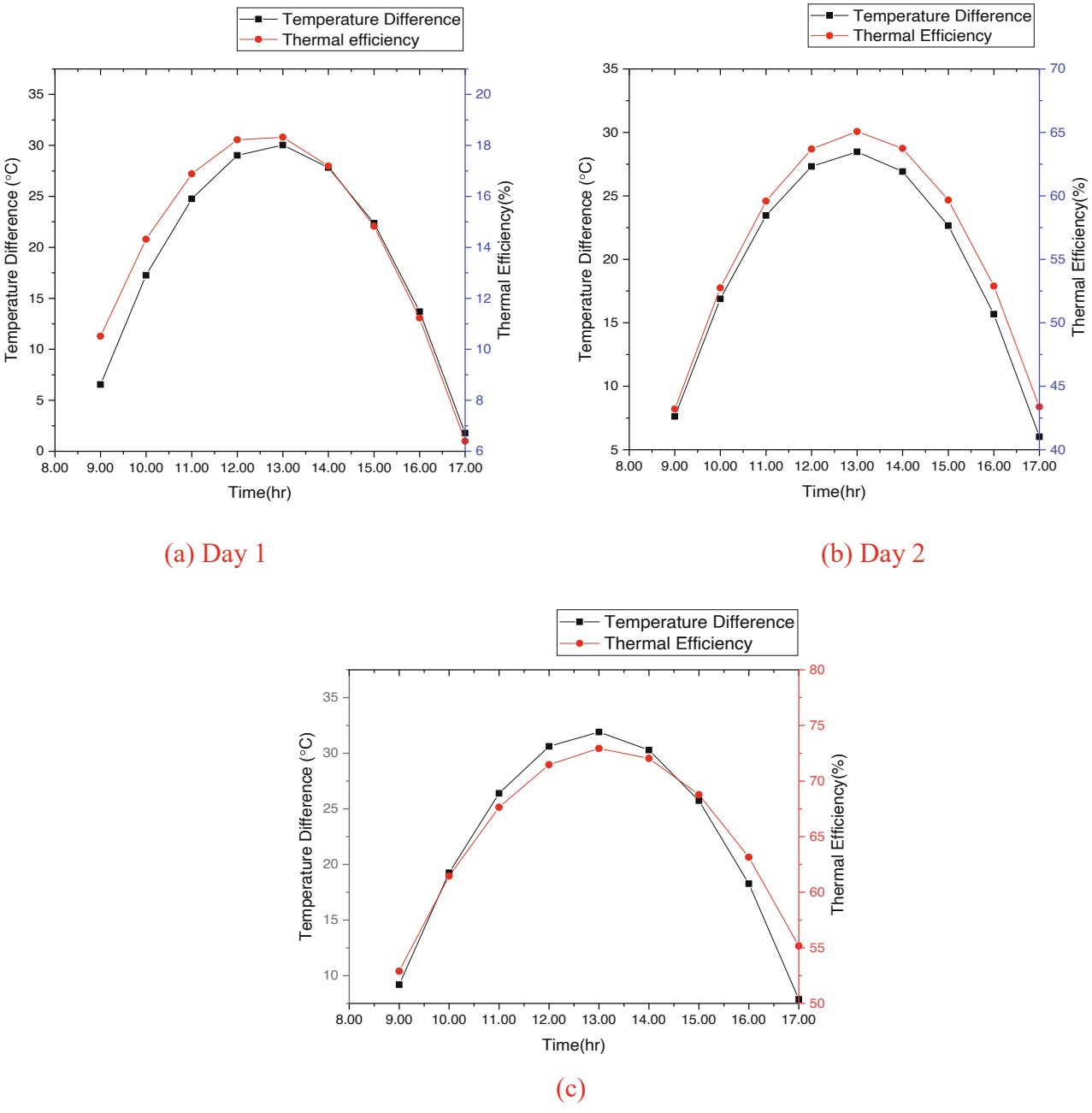


Fig. 7. (a-c): Comparison of temperature difference (black line) and thermal efficiency for time for three days of experiment.

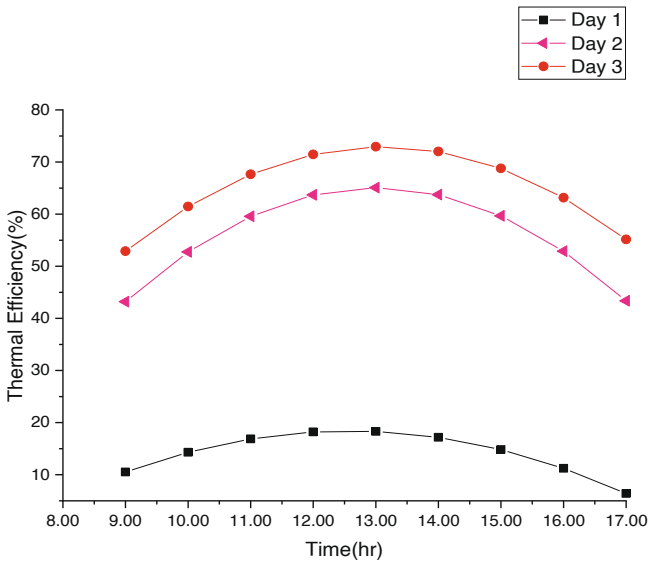


Fig. 8. Thermal Efficiency variation for three days of experiment.

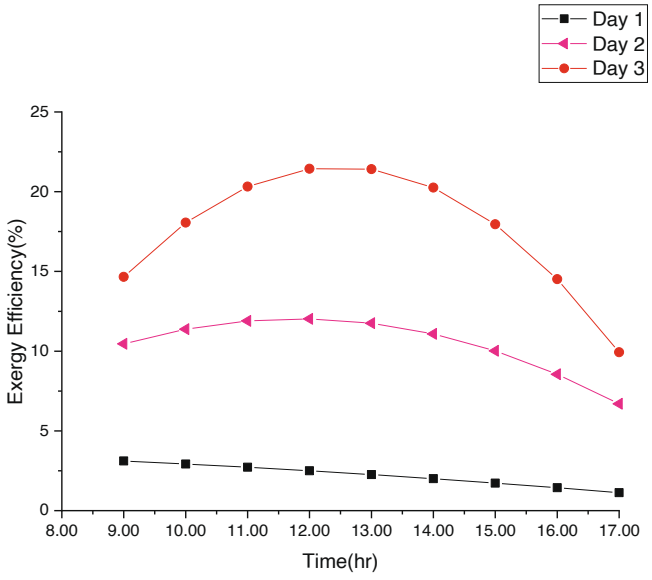


Fig. 9. Exergy efficiency variation for three days of experiment.

increases as the energy utilization temperature reaches inlet energy temperature.

Effect of reflector

Reflector harness solar energy and direct solar radiation to the SAH, leading to increased heat energy through the absorber plate. Reflector assembly was added on the third day of the experiment in induced forced convection mode. Average inlet and outlet temperature found was 32.93 °C and 55.32 °C, respectively. Average thermal and exergy efficiency was evaluated as 65.05% and 16.17%, respectively. Figs. 7 and 8 show the comparison of thermal and exergy efficiency in induced forced convection mode with and without the reflector. Hence the use of reflector with the SAH is fully justified.

Validation

Table 4 presented a comparative study about multi-pass and different solar air heating systems and proposed system designs. Proposed system having thermal efficiency of up to 86.19%; however, double pass air heater with fins cited in Table 4 have thermal efficiency in the range of 61–93%. Therefore, thermal efficiency of solar air heating collector can be enhanced without fins by implementation of reflector and thermal energy storage systems.

Table 4
Summary of selected study about multi-pass and different designs of solar air heating systems.

S. No.	Type	Additional	Thermal Efficiency	Ref.
1	Double Pass	Fins	73.5%	[50]
2	Double Pass	Tubular absorber	86.03%	[51]
3	Double Pass	Finned and wire mesh	93%	[19]
4	Double Pass	Corrugated plate and wire mesh	80%	[52]
5	Double Pass	V-groove absorber	61–77%	[53]
6	Double pass	Aluminum canes	<67%	[31]
7	Double Pass	Mesh absorber	<80.39%	[54]
8	Triple-pass	Thermal energy storage	60%	[55]
9	Triple pass	V groove	<76.39%	[56]
10	Triple-pass	–	67%	[57]
11	Triple-pass and quadruple-pass	V-groove absorber	<91.9%	[58]
12	Quadruple-pass	–	<90.3%	[56]
13	Tube-type	Turbulators	<72.41	[59]
14	Tube-type	Iron mesh modification	<67.69%	[60]
15	Parallel-pass	Plus-shaped perforated baffles	<75.11%	[61]
16	**Double pass**	**Sensible heat storage, Metico coated absorber with reflector**	**<86.19%**	**Present study**

Conclusions

In this work, an innovative DPSAHSHC with reflector is designed and developed and experimentally tested in three consecutive days under similar weather conditions. The main outcomes of this study are summarized as follow:

- A difference in solar radiation is observed for the day, but the difference in each day's daily average solar radiation is comparatively less.
- Outlet temperature of the air flowing out of the DPSAHSHC directly influences incident solar radiation. Average outlet air temperature of 3rd day experiment is 7.52 °C more than that of 2nd day of the experiment, while the inlet air temperature remained almost same. Reflector was used on 3rd-day of the experiment diverted more solar energy on the absorber plate, thus increasing the outlet air temperature.
- Rise in air temperature inside SAH has a direct influence on thermal efficiency. Average thermal efficiency observed was 65.05% and 56% on 3rd and 2nd day. Hence, an increase of 9% thermal efficiency was observed with a temperature difference of 7.52 °C.
- Proposed system has better exergy and thermal efficiency in a bottom outlet than natural convective mode. MFR on 1st day was 0.005 kg/s without dc fan, while on 2nd day the MFR observed was 0.02 kg/s, which is 0.015 kg/s more than 1st day. Average thermal efficiency was 1.79% more in induced forced convection than the natural convection. Exergic efficiency increased by 10% on the 2nd day in comparison to 1st day. The dc fan provides more airflow and reducing heat loss leads to better thermal and exergy performance. Outlet temperature decreases when the airflow rate expands, whereas when MFR increases, the collector efficiency improves, but it causes higher energy extraction. SAH utilizes dark painted material to increment its productivity.

- Reflector in the bottom outlet in induced forced mode SAH had better thermal and exergy performance. Thermal efficiency increased by 9% on 3rd day than 2nd day despite having the same 0.02 kg/s MFR. Likewise, the exergy efficiency was 2.96% more on 3rd day than that on 2nd day.

Experimental and thermal results show the successful design of the proposed system. This system can be used in various low and medium temperature applications such as space heating, desalination, drying, and other industrial and domestic application.

CRediT authorship contribution statement

Om Prakash: Formal analysis, Resources, Validation, Visualization, Writing – review & editing. **Anil Kumar:** Conceptualization, Project administration, Writing – review & editing. **Samsher:** Data curation, Supervision. **Kumaresh Dey:** Investigation. **Ankish Aman:** Methodology, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Kareem MW, Habib K, Sulaiman SA. Comparative study of single pass collector and double pass solar collector filled with porous media. Asian J Sci Res 2013;6(3):445–55.
[2] Bahrehmand D, Ameri M. Energy and exergy analysis of different solar air collector systems with natural convection. Renewable Energy 2015;74:357–68.
[3] Velmurugan P, Kalaivanan R. Energy and exergy analysis of multi-pass flat plate solar air heater—an analytical approach. Int J Green Energy 2015;12(8):810–20.
[4] Ghiami A, Ghiami S. Comparative study based on energy and exergy analyses of a baffled solar air heater with latent storage collector. Appl Therm Eng 2018;133:797–808.
[5] Chamoli S, Thakur NS. Exergetic performance evaluation of solar air heater having V-down perforated baffles on the absorber plate. J Therm Anal Calorim 2014;117(2):909–23.
[6] El Khadraoui A, Bouadila S, Kooli S, Guizani A, Farhat A. Solar air heater with phase change material: An energy analysis and a comparative study. Appl Therm Eng 2016;107:1057–64.
[7] Chabane F, Moummi N, Benramache S. Experimental study of heat transfer and thermal performance with longitudinal fins of solar air heater. J Adv Res 2014;5(2):183–92.
[8] Zukowski M. Experimental investigations of thermal and flow characteristics of a novel microjet air solar heater. Appl Energy 2015;142:10–20.
[9] Saxena A, Srivastava G, Tirth V. Design and thermal performance evaluation of a novel solar air heater. Renewable Energy 2015;77:501–11.
[10] Siddiqui FR, Elminshawy NAS, Addas MF. Design and performance improvement of a solar desalination system by using solar air heater: Experimental and theoretical approach. Desalination 2016;399:78–87.
[11] Pramanik RN, Sahoo SS, Swain RK, Mohapatra TP, Srivastava AK. Performance analysis of double pass solar air heater with bottom extended surface. Energy Procedia 2017;109:331–7.
[12] Kalaiarasi G, Velraj R, Swami MV. Experimental energy and exergy analysis of a flat plate solar air heater with a new design of integrated sensible heat storage. Energy. 2016;15(111):609–19.
[13] Edalatpour M, Kianifar A, Aryana K, Tiwari GN. Energy, exergy, and cost analyses of a double-glazed solar air heater using phase change material. J Renewable Sustainable Energy 2016;8(1):015101.
[14] Fudholi A, Sopian K, Othman MY, Ruslan MH, Bakhtyar B. Energy analysis and improvement potential of finned double-pass solar collector. Energy Convers Manage 2013;75:234–40.
[15] Srinivas M, Jayaraj S. Energy and exergy analysis of a two pass photovoltaic-thermal (PV/T) air heater. Int J Energy Environ 2013;4:467–80.
[16] Abuṣka M. Energy and exergy analysis of solar air heater having new design absorber plate with conical surface. Appl Therm Eng 2018;131:115–24.
[17] Heydari A, Mesgarpour M. Experimental analysis and numerical modeling of solar air heater with helical flow path. Sol Energy 2018;162:278–88.
[18] Salih SM, Jalil JM, Najim SE. Experimental and numerical analysis of double-pass solar air heater utilizing multiple capsules PCM. Renewable Energy 2019;143:1053–66.
[19] Singh S, Dhruw L, Chander S. Experimental investigation of a double pass converging finned wire mesh packed bed solar air heater. J Storage Mater 2019;21:713–23.
[20] Rhee SJ, Edwards DK. Laminar entrance flow in a flat plate duct with asymmetric suction and heating. Num Heat Transf 1981;4(1):85–100.
[21] Prasad BN, Sah GN. Plate temperature and heat transfer characteristics of artificially roughened solar air heater. Energy Procedia 2014;62:256–69.
[22] Manjunath MS, Karanth KV, Sharma NY. Numerical analysis of the influence of spherical turbulence generators on heat transfer enhancement of flat plate solar air heater. Energy 2017;121:616–30.
[23] Thakur DS, Khan MK, Pathak M. Performance evaluation of solar air heater with novel hyperbolic rib geometry. Renewable Energy 2017;1(105):786–97.
[24] Jaurker AR, Saini JS, Gandhi BK. Heat transfer and friction characteristics of rectangular solar air heater duct using rib-grooved artificial roughness. Sol Energy 2006;80(8):895–907.
[25] Singh S, Dhruw L, Chander S. Experimental investigation of a double pass converging finned wire mesh packed bed solar air heater. J Storage Mater 2019;1(21):713–23.
[26] Tyagi RK, Ranjan R, Kishore K. Performance studies on flat plate solar air heater subjected to various flow patterns. Appl Solar Energy 2014;50(2):98–102.
[27] Pfister H, Ralston T, Kim SW. A novel gridded solar air heater and an investigation of its conversion efficiency. Sol Energy 2016;136:560–70.
[28] Krishnananth SS, Kalidasa Murugavel K. Experimental study on double pass solar air heater with thermal energy storage. J King Saud Univ-Eng Sci 2013;25(2):135–40.
[29] Kumar A, Layek A. Nusselt number and friction factor correlation of solar air heater having twisted-rib roughness on absorber plate. Renewable Energy 2019;130:687–99.
[30] Baissi MT, Brima A, Aoues K, Khanniche R, Moummi N. Thermal behavior in a solar air heater channel roughened with delta-shaped vortex generators. Appl Therm Eng 2020;165:113563.
[31] Ozgen F, Esen M, Esen H. Experimental investigation of thermal performance of a double-flow solar air heater having aluminium cans. Renewable Energy 2009;34(11):2391–8.
[32] Jha RK, Choudhury C, Garg HP, Zaidi ZH. Performance prediction of a solar heated house. Energy Convers Manage 1992;33(4):263–73.
[33] Kumar R, Goel V, Kumar A. Investigation of heat transfer augmentation and friction factor in triangular duct solar air heater due to forward facing chamfered rectangular ribs: A CFD based analysis. Renewable Energy 2018;115:824–35.
[34] Jia B, Liu F, Wang Da. Experimental study on the performance of spiral solar air heater. Sol Energy 2019;182:16–21.
[35] Nidhul K, Yadav AK, Anish S, Arunachala UC. Efficient design of an artificially roughened solar air heater with semi-cylindrical side walls: CFD and exergy analysis. Sol Energy 2020;207:289–304.
[36] Esen H. Experimental energy and exergy analysis of a double-flow solar air heater having different obstacles on absorber plates. Build Environ 2008;43(6):1046–54.
[37] Bouadila S, Lazaar M, Skouri S, Kooli S, Farhat A. Energy and exergy analysis of a new solar air heater with latent storage energy. Int J Hydrogen Energy 2014;39(27):15266–74.
[38] Kurtbas İ, Durmuş A. Efficiency and exergy analysis of a new solar air heater. Renewable Energy 2004;29(9):1489–501.
[39] Abuṣka M. Energy and exergy analysis of solar air heater having new design absorber plate with conical surface. Appl Therm Eng 2018;131:115–24.
[40] Acır A, Canlı ME, Ata İ, Çakıroğlu R. Parametric optimization of energy and exergy analyses of a novel solar air heater with grey relational analysis. Appl Therm Eng 2017;122:330–8.
[41] Matheswaran MM, Arjunan TV, Somasundaram D. Analytical investigation of solar air heater with jet impingement using energy and exergy analysis. Sol Energy 2018;161:25–37.
[42] Alta D, Bilgili E, Ertekin C, Yaldiz O. Experimental investigation of three different solar air heaters: Energy and exergy analyses. Appl Energy 2010;87(10):2953–73.
[43] Ghaderian J, Azwadi CN, Mohammed H. Modelling of energy and exergy analysis for a double-pass solar air heater system. J Adv Res Fluid Mech Therm Sci 2015;16:15–32.
[44] Koroneos C, Nanaki E, Xydis G. Solar air conditioning systems and their applicability—an exergy approach. Resour Conserv Recycl 2010;55(1):74–82.
[45] Kumar A, Saini RP, Saini JS. A review of thermohydraulic performance of artificially roughened solar air heaters. Renew Sustain Energy Rev 2014;1(37):100–22.
[46] Prakash O, Kumar A, editors. Solar drying technology: concept, design, testing, modeling, economics, and environment. Springer 2017;29.
[47] Menni Y, Azzi A, Chamkha A. The solar air channels: comparative analysis, introduction of arc-shaped fins to improve the thermal transfer. J Appl Comput Mech 2019;5(4):616–26.
[48] Öztürk HH. Experimental evaluation of energy and exergy efficiency of a seasonal latent heat storage system for greenhouse heating. Energy Convers Manage 2005;46(9-10):1523–42.
[49] Minh Phu N, Thanh Luan N. A review of energy and exergy analyses of a roughened solar air heater. J Adv Res Fluid Mech Therm Sci 2020;77(2):160–75.

[50] Salih MMM, Alomar OR, Ali FA, Abd HM. An experimental investigation of a double pass solar air heater performance: A comparison between natural and forced air circulation processes. Sol Energy 2019;193:184–94.
[51] Abo-Elfadl S, Hassan H, El-Dosoky MF. Study of the performance of double pass solar air heater of a new designed absorber: An experimental work. Sol Energy 2020;198:479–89.
[52] Singh S. Experimental and numerical investigations of a single and double pass porous serpentine wavy wiremesh packed bed solar air heater. Renewable Energy 2020;145:1361–87.
[53] Bashria A, Yousef A, Adam NM, Sopian K, Zaharim A, Alghoul M. Analysis of single and double passes v-grooves solar collector with and without porous media. Int J Energy Environ 2007;2:109–14.
[54] Güler HÖzge, Sözen A, Tuncer ADoğuş, Afshari F, Khanlari A, Şirin C, et al. Experimental and CFD survey of indirect solar dryer modified with low-cost iron mesh. Sol Energy 2020;197:371–84.
[55] Kesavan S, Arjunan TV. Experimental study on triple pass solar air heater with thermal energy storage for drying mint leaves. Energy Technol Policy 2018;14(1):34–48.
[56] Tuncer ADoğuş, Sözen A, Khanlari A, Amini A, Şirin C. Thermal performance analysis of a quadruple-pass solar air collector assisted pilot-scale greenhouse dryer. Sol Energy 2020;203:304–16.
[57] Orbegoso EM, Saavedra R, Madrid MD, RL.. Numerical characterization of one-step and three-step solar air heating collectors used for cocoa bean solar drying. Environ Manage 2017;203:1080–94.
[58] Khanlari A, Sözen A, Afshari F, Şirin C, Tuncer ADoğuş, Gungor A. Drying municipal sewage sludge with v-groove triple-pass and quadruple-pass solar air heaters along with testing of a solar absorber drying chamber. Sci Total Environ 2020;709:136198.
[59] Afshari F, Sözen A, Khanlari A, Tuncer ADoğuş, Şirin C. Effect of turbulator modifications on the thermal performance of cost-effective alternative solar air heater. Renewable Energy 2020;158:297–310.
[60] Sözen A, Şirin C, Khanlari A, Tuncer AD, Gürbüz EY. Thermal performance enhancement of tube-type alternative indirect solar dryer with iron mesh modification. Sol Energy 2020;207:1269–81.
[61] Khanlari A, Güler HÖzge, Tuncer ADoğuş, Şirin C, Bilge YaşarC, Yılmaz Y, et al. Experimental and numerical study of the effect of integrating plus-shaped perforated baffles to solar air collector in drying application. Renewable Energy 2020;145:1677–92.